Exoplanet Hunters: Analysis of the Transit Data of STAC-1, STAC-2 and STAC-3

STAC WARS – Problem Statement 3 (PS 3)

Analysis Report

***Abstract*—In this report we study the brightness readings of three stars, STAC-1, STAC-2 and STAC-3, to look for planets passing in front of them. From the repeated dips in brightness, we find the orbital period and transit depth of one planet around each star. Using the transit depth, we estimate the radius of every planet, and using Kepler's third law we estimate its distance from the star. Finally we compare the three planets with the simplified habitable-zone distance given in the problem and pick the most promising one. We find that the planet around STAC-3 is the best candidate for further study, although our data cannot show that it is habitable. One unexplained dip in the STAC-3 data is also discussed.**

***Index Terms*—exoplanets, transit, transit depth, Kepler's third law, habitable zone**

I. Introduction

We are given brightness readings of three stars, STAC-1, STAC-2 and STAC-3, recorded by a space telescope over time.

When an exoplanet crosses its star, we see a dip in the Rel. Brightness, such event is called a transit.

Using the data, constants and formulae given in the problem statement, we try to:

(i) check whether each star shows changes in Rel. brightness,

(ii) estimate the size of each planet,

(iii) estimate how far each planet is from its star, and

(iv) choose the planet that is the best candidate for further habitability study.

II. Data and Method

A. Given Data

The properties of the three stars are listed in Table I. The constants we use are: solar radius $R_\odot$ = 696,000 km, Earth radius $R_\oplus$ = 6,371 km, 1 AU = 149.6 × 10^6 km and 1 year = 365.25 days.

TABLE I
STELLAR PROPERTIES GIVEN IN THE PROBLEM

Property	STAC-1	STAC-2	STAC-3
Mass ($M_\odot$)	1.00	0.80	0.50
Radius ($R_\odot$)	1.00	0.80	0.50
Luminosity ($L_\odot$)	1.00	0.40	0.08
Temperature (K)	5800	5000	3800

B. Assumptions

1) The orbits are circular, so a transit happens in every orbit.
2) We consider the planet as an opaque object and take its projected area upon the star considering it to be blocking the light.
3) The mass of the planet is very small compared to the mass of the star.
4) Every repeated dip in brightness is caused by a planet itself.
5) The problem does not give size limits, so we assume these approximate ranges: Earth-sized (below 1.25 $R_\oplus$), Super-Earth (1.25 to 3 $R_\oplus$), Neptune-sized (3 to 10 $R_\oplus$) and Jupiter-sized (above 10 $R_\oplus$).

C. Formulae Used

The transit depth is the fractional drop in brightness:

$$\delta = (F_{normal} - F_{transit}) / F_{normal} \quad (1)$$

The approximate planet radii can be calculated by :

$$\delta \approx (R_p / R_s)^2, \quad R_p = R_s\sqrt{\delta} \quad (2)$$

For the orbit we use Kepler's third law in the simplified form $P^2 = a^3/M$, where P is in years, a in AU and M in solar masses. Rearranging,

$$a = (M P^2)^{1/3} \quad (3)$$

The period in days is divided by 365.25 to get years. For the habitable-zone comparison the problem gives

$$d_{HZ} \approx \sqrt{(L / L_\odot)} \ \text{AU} \quad (4)$$

III. Results

A. Part A: Detecting the Exoplanets

STAC-1. The brightness drops from 1.0000 to 0.9970 at days 3, 8, 12, 16 and 20. The depth is $\delta = (1.0000 - 0.9970)/1.0000 = 0.0030$, i.e. 0.30%. The gaps between the dips are 5, 4, 4 and 4 days. The first gap is different from the others and we are not sure why. Three of the four gaps are 4 days, so we take P = 4 days. The mean gap is 4.25 days; we check in Part C how much this changes our answer.

STAC-2. The brightness drops to 0.9990 at days 20, 100 and 180, so $\delta = 0.0010$ (0.10%). The gaps are 80 and 80 days, so P = 80 days.

STAC-3. The brightness drops at days 20, 60, 100 and 140, with a gap of 40 days each time, so P = 40 days. The depths are 0.0010, 0.0008, 0.0009 and 0.0007. These are not exactly equal and we don't have any explanation for it. We use the largest value, $\delta = 0.0010$, as the transit depth. There is one more dip of 0.0010 at day 170, which does not fit the 40-day pattern, and there is no dip at day 180 where the next transit should be. We could not explain this dip with the given data, so we do not use it in our calculations. (A hypothesis is that it may be some error in the equipment with which the reading has been taken, or the fact that the assumption made by us that the decrease in Rel. Brightness is only caused by exoplanets may be wrong.)

Why a repeating dip suggests a planet:

A planet orbiting a star passes in front of it once every orbit and blocks a small part of its light. So the brightness falls by about the same amount again and again after equal time gaps. A dip that repeats like this suggests that something is orbiting

the star. All three stars show this behaviour, so we call the planets STAC-1 p, STAC-2 p and STAC-3 p.

B. Part B: Planet Sizes

We use $R_p = R_s\sqrt{\delta}$ from (2) with R_s = (radius in $R_\odot$) × 696,000 km.

STAC-1 p: R_s = 696,000 km

$$R_p = 696{,}000 \times \sqrt{0.0030} = 696{,}000 \times 0.05477 \approx 38{,}121 \text{ km}$$
$$R_p/R_\oplus = 38{,}121 / 6{,}371 \approx 5.98$$

STAC-2 p: $R_s = 0.80 \times 696{,}000 = 556{,}800$ km

$$R_p = 556{,}800 \times \sqrt{0.0010} = 556{,}800 \times 0.03162 \approx 17{,}608 \text{ km}$$
$$R_p/R_\oplus = 17{,}608 / 6{,}371 \approx 2.76$$

STAC-3 p: $R_s = 0.50 \times 696{,}000 = 348{,}000$ km

$$R_p = 348{,}000 \times \sqrt{0.0010} = 348{,}000 \times 0.03162 \approx 11{,}005 \text{ km}$$
$$R_p/R_\oplus = 11{,}005 / 6{,}371 \approx 1.73$$

Using our assumed ranges, STAC-1 p is Neptune-sized, and STAC-2 p and STAC-3 p are Super-Earths. The results are collected in Table II.

TABLE II
TRANSIT RESULTS AND PLANET SIZES

Planet	P (d)	δ	R_p (km)	$R_p/R_\oplus$	Class
STAC-1 p	4	0.0030	38,121	5.98	Neptune-sized
STAC-2 p	80	0.0010	17,608	2.76	Super-Earth
STAC-3 p	40	0.0010	11,005	1.73	Super-Earth

C. Part C: Planetary Orbits

We convert P to years and use $a = (MP^2)^{1/3}$ from (3).

STAC-1 b: P = 4/365.25 = 0.010951 yr, M = 1.00

$$a = (1.00 \times 0.010951^2)^{1/3} \approx 0.0493 \text{ AU}$$
$$a = 0.0493 \times 149.6 \times 10^6 \approx 7.38 \times 10^6 \text{ km}$$

STAC-2 p: P = 80/365.25 = 0.21903 yr, M = 0.80

$$a = (0.80 \times 0.21903^2)^{1/3} \approx 0.3373 \text{ AU}$$
$$a = 0.3373 \times 149.6 \times 10^6 \approx 5.05 \times 10^7 \text{ km}$$

STAC-3 p: P = 40/365.25 = 0.10951 yr, M = 0.50

$$a = (0.50 \times 0.10951^2)^{1/3} \approx 0.1817 \text{ AU}$$
$$a = 0.1817 \times 149.6 \times 10^6 \approx 2.72 \times 10^7 \text{ km}$$

If we use P = 4.25 days for STAC-1 p instead, we get a ≈ 0.0513 AU, a change of only about 4%, so this choice does not matter much. The distances are collected in Table III.

TABLE III
ORBITAL DISTANCES OF THE PLANETS

Planet	P (d)	M ($M_\odot$)	a (AU)	a (km)
STAC-1 p	4	1.00	0.0493	7.38×10^6
STAC-2 p	80	0.80	0.3373	5.05×10^7
STAC-3 p	40	0.50	0.1817	2.72×10^7

Comparison with Earth. Earth is 1 AU from the Sun. STAC-1 p is at about 1/20 of this distance, STAC-2 p at about 1/3 and STAC-3 p at about 1/5.5. All three planets are much closer to their stars than Earth is to the Sun.

D. Part D: Most Promising Planet

We use $d_{HZ} = \sqrt{(L/L_\odot)}$ from (4):

STAC-1 p: $d_{HZ} = \sqrt{1.00} = 1.000$ AU, $a/d_{HZ} = 0.0493$
STAC-2 p: $d_{HZ} = \sqrt{0.40} = 0.632$ AU, $a/d_{HZ} = 0.533$
STAC-3 p: $d_{HZ} = \sqrt{0.08} = 0.283$ AU, $a/d_{HZ} = 0.642$

Since $d_{HZ}^2 = L/L_\odot$, a planet at d_{HZ} receives the same starlight as Earth gets from the Sun. Starlight spreads out as $1/a^2$, so compared to Earth a planet receives

$$S = (L/L_\odot) / a^2 = (d_{HZ} / a)^2 \quad \text{(Earth = 1)} \qquad (5)$$

This gives S ≈ 411 for STAC-1 p, 3.5 for STAC-2 p and 2.4 for STAC-3 p. The values are in Table IV.

TABLE IV
HABITABLE-ZONE COMPARISON

Planet	d_{HZ} (AU)	a (AU)	a/d_{HZ}	Starlight S
STAC-1 p	1.000	0.0493	0.0493	≈ 411
STAC-2 p	0.632	0.3373	0.533	≈ 3.5
STAC-3 p	0.283	0.1817	0.642	≈ 2.4

The problem does not say how to rank the planets, so we use the following reasoning: the closer a planet is to d_{HZ}, the closer the starlight it receives is to what Earth gets. All three planets have $a < d_{HZ}$, so all get more starlight than Earth, and none is inside the habitable-zone distance. The planet with the largest a/d_{HZ} is therefore the least overheated and the best candidate. This is STAC-3 p. Our evidence is:

1) **Distance:** STAC-3 p has the largest a/d_{HZ} (0.642), compared to 0.533 for STAC-2 p and 0.0493 for STAC-1 p. It is the closest to the habitable-zone distance, while STAC-1 p is about twenty times too close.
2) **Size:** its radius of about 1.73 $R_\oplus$ is the smallest of the three planets (2.76 $R_\oplus$ and 5.98 $R_\oplus$ for the others), so it is the closest to Earth's size. Our data cannot tell what it is made of.
3) **Starlight:** it receives only about 2.4 times Earth's starlight, compared to 3.5 times for STAC-2 p and 411 times for STAC-1 p. It is the least heated of the three.

IV. DISCUSSION

The results depend on a few choices which we are not fully sure about. For STAC-1 the first gap between dips is 5 days while the others are 4 days. We could not find the reason from the data, so we used the 4-day period that fits most of the gaps. The other choice (4.25 days) changes the distance by only about 4%, so we do not think it affects the conclusion. STAC-2 shows only three dips, so its period should be checked with more observations.

STAC-3 is the least clear system. The four regular dips have slightly different depths, and there is an extra dip at day 170 that does not fit the 40-day pattern. We do not know if this dip is a second planet, a measurement error or something else, and one dip alone is not enough to say. We used the deepest regular dip, 0.0010, for the size of STAC-3 p. If the shallower depths were used instead, the radius would be a bit smaller, but it would still be a Super-Earth and the ranking in Part D would not change, because that ranking depends only on the period, the mass and the luminosity.

Our choice of STAC-3 p is only a comparison between three planets. STAC-2 p is not far behind it (0.533 against 0.642), and

all three planets receive more starlight than Earth. Being near the habitable-zone distance also does not mean a planet is habitable, because we know nothing about its atmosphere or surface. More observations of the STAC-3 star, especially around day 180 and later, would help to explain the day-170 dip.

V. Conclusion

All three stars show repeating dips in brightness, which suggests one planet around each star. The periods are 4, 80 and 40 days for STAC-1 p, STAC-2 p and STAC-3 p. Their radii are about 5.98, 2.76 and 1.73 Earth radii, so STAC-1 p is Neptune-sized and the other two are Super-Earths. They orbit at 0.0493, 0.3373 and 0.1817 AU, respectively. All much closer to their stars than Earth is to the Sun.

Comparing the planets with the habitable-zone distance, STAC-3 p is closest to it, has the smallest radius and receives the least starlight, so we choose it as the most promising candidate for further study. This does not prove it is habitable. Observations in different colours of light during a transit, and more brightness readings of the STAC-3 star, would be the next steps.